

MATH 301: Homework 5

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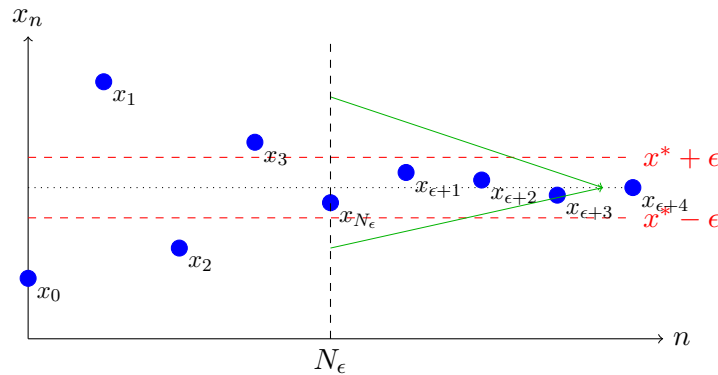
Cauchy Definition A sequence $X = (x_n)$ is called a *Cauchy sequence* if the terms of the sequence get arbitrarily close to each other as the sequence progresses. More formally,

For every $\epsilon > 0$, there exists an integer $N_\epsilon > 0$, such that

$$\forall m, n \geq N_\epsilon, \quad |x_n - x_m| < \epsilon.$$

This means that after some index N_ϵ , the distance between any two terms x_n and x_m (with n, m large enough) is less than any chosen positive number ϵ . Intuitively, the sequence becomes "clustered" together and the terms never drift far apart, no matter how far out in the sequence you go.

Example of a Cauchy Sequence



Section 3.5: 2b) Show directly from the definition that the following is a Cauchy sequence:

$$x_n := \left(1 + \frac{1}{2!} + \cdots + \frac{1}{n!} \right)$$

"English": x_n is Cauchy if there's some index N_ϵ s.t. given any two indices m, k where $N_\epsilon \leq m, k$. The differences of the two indices is "sufficiently small": $|x_m - x_k| < \epsilon$

Define terms:

$$x_m := \sum_{i=m \mid N_\epsilon < m}^{\infty} \frac{1}{m!} \quad x_k := \sum_{i=k \mid N_\epsilon < k}^{\infty} \frac{1}{n!}$$

Goal (Show the Following):

$$\forall \epsilon > 0, \quad \left| \sum_{i=m \mid N_\epsilon < m}^{\infty} \frac{1}{m!} - \sum_{i=k \mid N_\epsilon < k}^{\infty} \frac{1}{n!} \right| < \epsilon$$

$$\equiv \left| 1 + \frac{1}{2!} + \frac{1}{m!} - 1 - \frac{1}{2!} - \frac{1}{n!} \right| < \epsilon$$

Suppose $n < m$:

$$\equiv \left| \frac{1}{(n+1)!} + \cdots + \frac{1}{m!} \right| \leq \left| \frac{1}{2^{n+1}} + \frac{1}{2^{n+2}} + \cdots + \frac{1}{2^m} \right| \equiv \sum_{k=n+1}^{\infty} \left(\frac{1}{2} \right)^k$$

$$\sum_{k=n+1}^{\infty} \left(\frac{1}{2} \right)^k = \frac{1}{2^{m-n}}$$

$$\frac{1}{2^{m-n}}$$

[Section 3.5: 3c](#)) Show directly from the definition that the following is NOT a Cauchy sequence:

$$\ln(n)$$

[Section 3.5: 4](#) Show directly from the definition that if (x_n) and (y_n) are Cauchy sequences, then $(x_n + y_n)$ and $(x_n y_n)$ are Cauchy sequences

[Section 3.5: 7](#) Let (x_n) be a Cauchy sequence such that x_n is an integer for every $n \in \mathbb{N}$. Show that (x_n) is ultimately constant.